A0 ENGINEERING DRAFT — net labels connect equal names; NC crosses mark intentionally unused pins

Native USB / Ethernet / OLED need new firmware. PCB is deliberately unrouted. No verified electrical ratings yet.

POWER
J1 24V IN / GND
VIN24
GND
F1 500mA / 60V
VIN_FUSED
D1 SS16
V24
U2 TSR 1-2433
VIN
3V3
+3V3
C1 22uF / 50V
C2 100nF / 50V
C3 10uF / 10V
U1 STM32F446RET6
VBAT
VDD
VDDA
VSS
VSSA
NRST
BOOT0
OSC_IN
OSC_OUT
PH0
PH1
PD2
PC0 PC1 PC2 PC3 PC4 PC5 PC6 PC7 PC8 PC9 PC10 PC11 PC12 PC13 PC14 PC15
BTN_N
VCAP1
VCAP_1
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9 PA10 PA11 PA12 PA13 PA14 PA15
PB0 PB1 PB2 PB3 PB4 PB5 PB6 PB7 PB8 PB9 PB10 PB12 PB13 PB14 PB15
DI1_N
DI2_N
UART_TX
UART_RX
DI3_N
STATUS_LED
DI4_N
ETH_INT_N
USB_VBUS_SENSE
ETH_RST_N
USB_DM
USB_DP
SWDIO
SWCLK
DO1
DO2
BOOT1
I2C_SCL
I2C_SDA
DO3
DO4
ETH_CS_N
SPI2_SCK
SPI2_MISO
SPI2_MOSI
MCU
C4 100nF / 50V
C5 100nF / 50V
C6 100nF / 50V
C7 100nF / 50V
C8 4.7uF / 10V
C9 4.7uF / 10V
R1 22
C10 100nF / 50V
C11 1uF / 10V
R2 10k
C12 100nF / 50V
R3 10k
R4 10k
J2 BOOT: 1=3V3 2=BOOT0 3=GND
J3 SWD: 3V3 GND DIO CLK NRST SWO
J4 UART: GND TX RX
Y1 8MHz CL18pF
C13 27pF C0G / 50V
C14 27pF C0G / 50V
SW1 RESET
SW2 USER
R5 10k
D2 GREEN
LED_A
R6 1k
USB
J13 USB-C DEVICE
VBUS
CC1
CC2
D−
D+
SBU1
SBU2
SHIELD
SH
VBUS_USB
USB_DM_CONN
USB_DP_CONN
R60 5.1k
R61 5.1k
R62 22
R63 22
R64 68k
R65 100k
U8 USBLC6-2SC6
C31 100nF / 50V
TEST POINTS
TP1 GND
TP2 +3V3
TP3 V24
TP4 VCAP1
INPUT 1
J5 DI1 / GND
DI1_24
R10 2.2k
DI1_MID
R11 2.2k
DI1_A
U4 VO617A-3
D11 1N4148W
R12 10k
C21 10nF / 50V
INPUT 2
J6 DI2 / GND
R20 2.2k
R21 2.2k
U5 VO617A-3
D12 1N4148W
R22 10k
C22 10nF / 50V
INPUT 3
J7 DI3 / GND
R30 2.2k
R31 2.2k
U6 VO617A-3
D13 1N4148W
R32 10k
C23 10nF / 50V
INPUT 4
J8 DI4 / GND
R40 2.2k
R41 2.2k
U7 VO617A-3
D14 1N4148W
R42 10k
C24 10nF / 50V
OUTPUTS
U3 TPL7407LDR
IN1 IN2 IN3 IN4 IN5 IN6 IN7
COM
OUT1 OUT2 OUT3 OUT4 OUT5 OUT6 OUT7
Q1 Q2 Q3 Q4
C30 100nF / 50V
J9 +24 / Q1 sink
J10 +24 / Q2 sink
J11 +24 / Q3 sink
J12 +24 / Q4 sink
R51 100k
R52 100k
R53 100k
R54 100k
EXPANSION
J14 SPI: 3V3 GND SCK MISO MOSI CS
J15 ETH: GND RST INT
R70 10k
R71 10k
R72 10k
R73 4.7k
R74 4.7k
J16 OLED: 3V3 GND SCL SDA
MOUNTING
H1 M3 NPTH
H2 M3 NPTH
H3 M3 NPTH
H4 M3 NPTH
PWR_FLAG
GND
V24
VDDA
VIN24
VIN_FUSED

UNROUTED / NOT FOR FABRICATION — see hardware/prototype-a/README.md

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